Matrices CHEAT SHEET

A MATRIX

$$\begin{matrix} & \text{coffee} & \text{cake} & \text{sandwich} \\ \text{Ana} \\ \text{Ben} \\ \text{Kai} \end{matrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \\ 8 \end{bmatrix}$$

orders × prices = bills
($3, $4, $5: our example)
each bill: a row × the prices

order $m \times n$: m rows, n columns (rows first)
a_{ij}: the entry in row i, column j; $a_{23} = 1$

TYPES AND EQUALITY

row $1 \times n$ · column $m \times 1$ · square $n \times n$
diagonal: 0 off the main diagonal · scalar: equal diagonal
identity I: 1 on the diagonal · zero O: all 0
upper / lower triangular: 0 below / above it
$A = B \iff$ same order and every $a_{ij} = b_{ij}$

$$\begin{bmatrix} x+y & 2 \\ 5 & x-y \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 2 \end{bmatrix} \Rightarrow x = 4, y = 2$$

ADDING AND SCALING

$(A \pm B)_{ij} = a_{ij} \pm b_{ij}$ (same order only)

$$\begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 4 & -1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix}$$

kA: every entry × k $\quad 2A - 3B = \begin{bmatrix} -10 & 9 \\ 4 & -15 \end{bmatrix}$

$A + B = B + A \quad (A + B) + C = A + (B + C)$
$A + O = A \quad A + (-A) = O \quad k(A + B) = kA + kB$

$2X + A = B \Rightarrow X = \frac{1}{2}(B - A) = \begin{bmatrix} 3/2 & -2 \\ -1 & 5/2 \end{bmatrix}$

MULTIPLICATION

$(AB)_{ij} =$ (row i of A) · (column j of B)
$(m \times n)(n \times p) = m \times p$: the inside sizes match

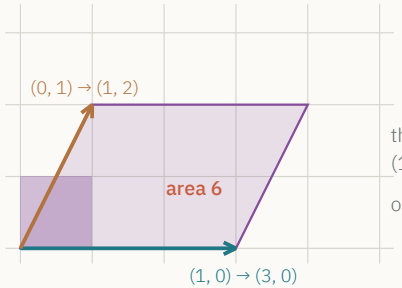
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix} \quad BA = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix} \neq AB$$

$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = O$: $AB = O$, $A \neq O$, $B \neq O$ (no cancelling)

$AI = IA = A \quad (AB)C = A(BC) \quad A(B + C) = AB + AC$

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \quad Q = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$: $Q^2 - 4Q + I = O$

A MATRIX MOVES THE PLANE



$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

the columns are where (1, 0) and (0, 1) land

origin fixed; lines stay lines

B first, then A: $x \to Bx \to A(Bx) = (AB)x$
turn then reflect ≠ reflect then turn

MATRICES OF MOVES

rotation by θ	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	det 1
reflection in the x axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	det −1
reflection in the y axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	det −1
reflection in y = x	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	det −1
enlargement, factor k	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$	det k^2
stretch along the x axis	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$	det k
shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	det 1

TRANSPOSE

$(A^T)_{ij} = a_{ji}$: rows become columns

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$(A^T)^T = A \quad (A + B)^T = A^T + B^T \quad (kA)^T = kA^T$
$(AB)^T = B^T A^T$ (the order reverses)

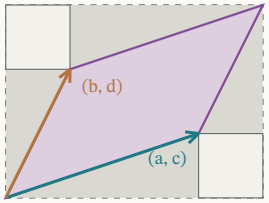
SYMMETRIC AND SKEW-SYMMETRIC

symmetric: $A^T = A$ $\quad$ skew-symmetric: $A^T = -A$
skew-symmetric ⇒ every diagonal entry is 0
$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

$$\begin{bmatrix} 1 & 5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Matrices CHEAT SHEET

THE DETERMINANT



$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

|det| = the area scale
negative: the shape is flipped
0: squashed flat, no inverse
rotation, shear: 1 · enlargement k: k^2

$(a + b)(c + d) - ac - bd - 2bc$

$\begin{vmatrix} 4 & 3 \\ 2 & 5 \end{vmatrix} = 20 - 6 = 14 \quad \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} = 0$

circle r = 1 by $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$: area $6\pi \approx 18.85$

3 × 3 DETERMINANTS

$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$ the chessboard of signs
expand along the line with most zeros
3 × 3: the volume scale

$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{vmatrix} = 1\cdot 24 - 2\cdot(-5) + 3\cdot(-4) = 22$

AREA OF A TRIANGLE

Area $= \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ (take its size)

(1, 1), (4, 2), (2, 5): $\frac{1}{2} \times 11 = 5.5$

$= 0 \Leftrightarrow$ collinear: (1, −1), (2, 1), (4, 5)

line through (1, 3), (2, 5): $\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 2 & 5 & 1 \end{vmatrix} = 0 \Rightarrow y = 2x + 1$

PROPERTIES OF DETERMINANTS

$|A^T| = |A|$

swap two rows: the sign changes

two equal rows: 0

one row × k: $|A| \times k$

$|kA| = k^n|A|$ (n × n)

$R_i \to R_i + kR_j$: no change

triangular: the product of the diagonal

$|AB| = |A|\,|B|$

MINORS AND COFACTORS

M_{ij}: cover row i and column j $\quad C_{ij} = (-1)^{i+j} M_{ij}$

$|A| = a_{i1}C_{i1} + a_{i2}C_{i2} + a_{i3}C_{i3}$ (any row i)

a row × another row's cofactors = 0

e.g. cofactors of $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{bmatrix}$

ADJOINT AND INVERSE

adj A = (matrix of cofactors)^T $\quad A\,(\text{adj } A) = |A|\,I$

$A^{-1} = \frac{\text{adj } A}{|A|}$ $(|A| \neq 0)$ $\quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}\begin{bmatrix} 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 25 \\ 67 \end{bmatrix}$

$(AB)^{-1} = B^{-1}A^{-1} \quad (A^T)^{-1} = (A^{-1})^T$

$|A^{-1}| = 1/|A| \quad |\text{adj } A| = |A|^{n-1}$

LINEAR SYSTEMS: AX = B

$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \\ 8 \end{bmatrix}$ $AX = B$

$X = A^{-1}B$ (multiply on the left) $\Rightarrow$ x, y, z = 3, 4, 5

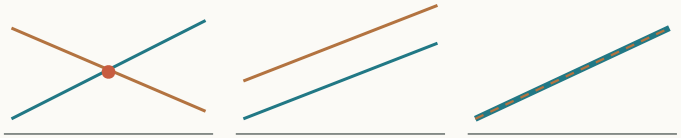
CRAMER'S RULE

$x = \frac{D_x}{D} \quad y = \frac{D_y}{D} \quad z = \frac{D_z}{D} \quad (D \neq 0)$

D_x: column 1 replaced by the constants

x + 2y = 8, 3x − y = 3: D = −7, D_x = −14, D_y = −21 $\Rightarrow$ (2, 3)

ONE, NONE OR INFINITELY MANY



cross: one | parallel: none | same line: infinitely many

|A| ≠ 0: exactly one solution
|A| = 0 and (adj A)B ≠ O: no solution
|A| = 0 and (adj A)B = O: infinitely many, or none
x + 2y = 4 with 2x + 4y = 8: one line · = 10: parallel